

Web Dev Mastery

✅ 📘 CSS Animations – Detailed Notes

CSS Animations allow elements to change their style or behavior over time without using JavaScript. You define **@keyframes** and apply them using animation properties.

🔷 1. `@keyframes`

- This is used to define the animation steps (how the element should change).
- You provide styles at certain points (like 0%, 50%, 100%).

css

```
@keyframes slideIn {
  from {
    transform: translateX(-100%);
  }
  to {
    transform: translateX(0);
  }
}
```

🔷 2. `animation-name`

- Specifies the name of the `@keyframes` to apply to the element.

css
```
animation-name: slideIn;
```

🔷 3. `animation-duration`

- Sets how long the animation should take (e.g., `2s`, `500ms`).

css
```
animation-duration: 2s;
```

🔷 4. `animation-delay`

- Adds a delay before the animation starts.

css
```
animation-delay: 1s;
```

🔷 5. `animation-iteration-count`

- Defines how many times the animation should repeat.
    - Values: `1`, `2`, `infinite`, etc.

css
```
animation-iteration-count: infinite;
```

🔷 6. `animation-direction`

- Specifies whether the animation should play forward, backward, or alternate.
    - `normal`: default
    - `reverse`: plays backwards
    - `alternate`: plays forward then backward
    - `alternate-reverse`: backward then forward

css
```
animation-direction: alternate;
```

🔷 7. `animation-timing-function`

- Controls the speed curve of the animation.
    - Common values: `linear`, `ease`, `ease-in`, `ease-out`, `ease-in-out`, `cubic-bezier(...)`

css
```
animation-timing-function: ease-in-out;
```

🔷 8. `animation-fill-mode`

- Defines what styles are applied to the element before/after the animation executes.
    - `none`: default
    - `forwards`: retains the end styles
    - `backwards`: applies styles at animation start

- `both`: applies both start and end styles

css
```
animation-fill-mode: forwards;
```

🔷 9. `animation` (Shorthand)

You can combine all the above properties in a single shorthand line:

css
```
animation: slideIn 2s ease-in-out 1s infinite alternate forwards;
```

✅ 🧪 Example: Simple Box Animation

html
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
  <title>CSS Animation Example</title>
  <style>
    .box {
      width: 100px;
      height: 100px;
      background-color: #3498db;
      margin: 50px auto;
      animation-name: moveBox;
      animation-duration: 3s;
      animation-delay: 1s;
      animation-iteration-count: infinite;
      animation-direction: alternate;
      animation-timing-function: ease-in-out;
      animation-fill-mode: forwards;
    }
```

```
    @keyframes moveBox {
      0% {
        transform: translateX(0);
        background-color: #3498db;
      }
      50% {
        transform: translateX(150px);
        background-color: #2ecc71;
      }
      100% {
        transform: translateX(300px);
        background-color: #e74c3c;
      }
    }
  </style>
</head>
<body>
  <div class="box"></div>
</body>
</html>
```

✅ 🔍 Summary Table

Property	Description
`@keyframes`	Defines the animation behavior
`animation-name`	Refers to the keyframe name
`animation-duration`	Time for one cycle of animation
`animation-delay`	Time to wait before animation starts
`animation-iteration-count`	Number of times the animation repeats
`animation-direction`	Direction of animation

`animation-timing-function`	Speed curve of animation
`animation-fill-mode`	Defines styles before/after animation
`animation`	Shorthand for all animation properties